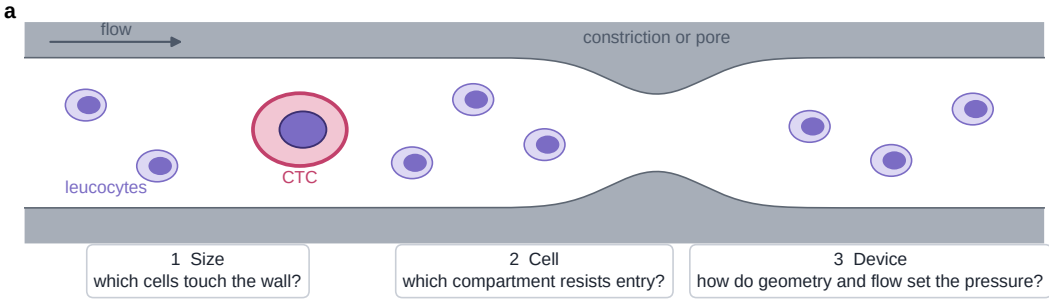
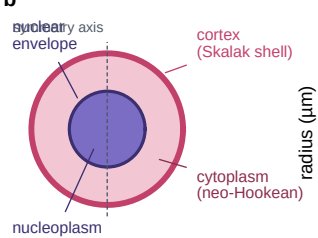
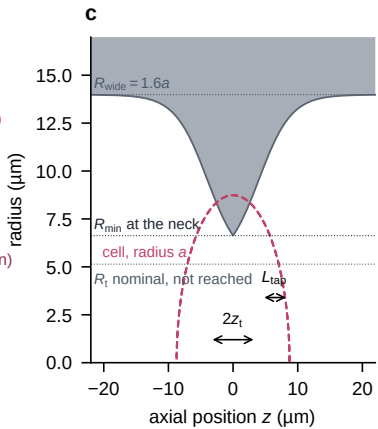
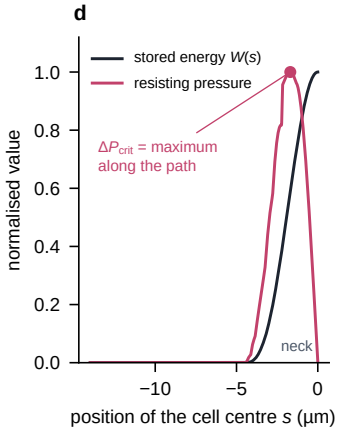

a
flow
constriction or pore
leucocytes
CTC
1 Size
which cells touch the wall?
2 Cell
which compartment resists entry?
3 Device
how do geometry and flow set the pressure?


b
nuclear envelope
cortex
(Skalak shell)
cytoplasm
(neo-Hookean)
nucleoplasm


c
$R_{wide} = 1.6a$
R_{min} at the neck
cell, radius a
R_t nominal, not reached
L_{tap}
$2z_t$
radius (µm)
axial position z (µm)


d
stored energy $W(s)$
resisting pressure
ΔP_{crit} = maximum
along the path
neck
normalised value
position of the cell centre s (µm)